

Security Assessment Report

Introduction

Security Assessment is the process of identifying vulnerabilities, misconfigurations, and security weaknesses in systems or networks. It helps organizations understand risks and improve their overall security posture.

Objective

The objective of this task was to learn how to perform a security assessment using open-source tools, identify vulnerabilities in a test environment, analyze the risks, and prepare a report with remediation recommendations.

Tools Used

Kali Linux was used as the testing machine.

VirtualBox was used to run virtual machines.

Metasploitable was used as the vulnerable target machine.

OpenVAS / Greenbone was used for vulnerability scanning.

Nmap was used for network scanning.

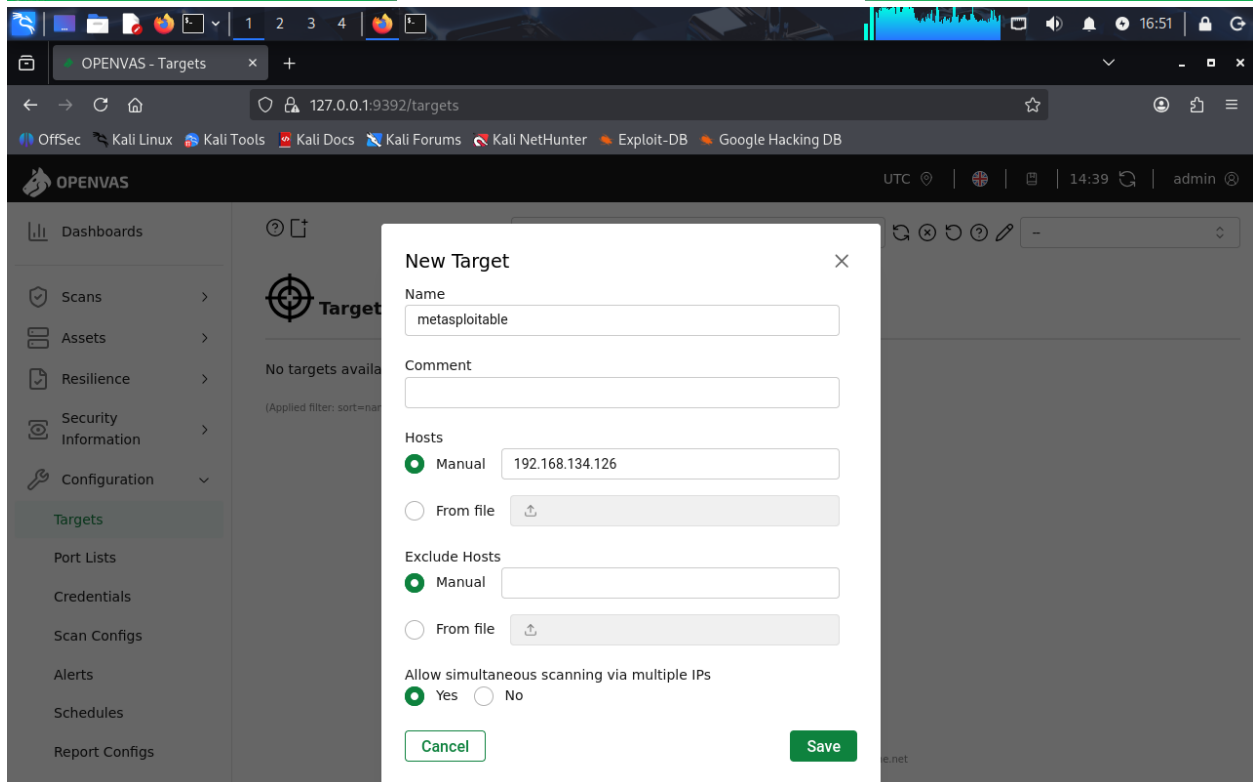
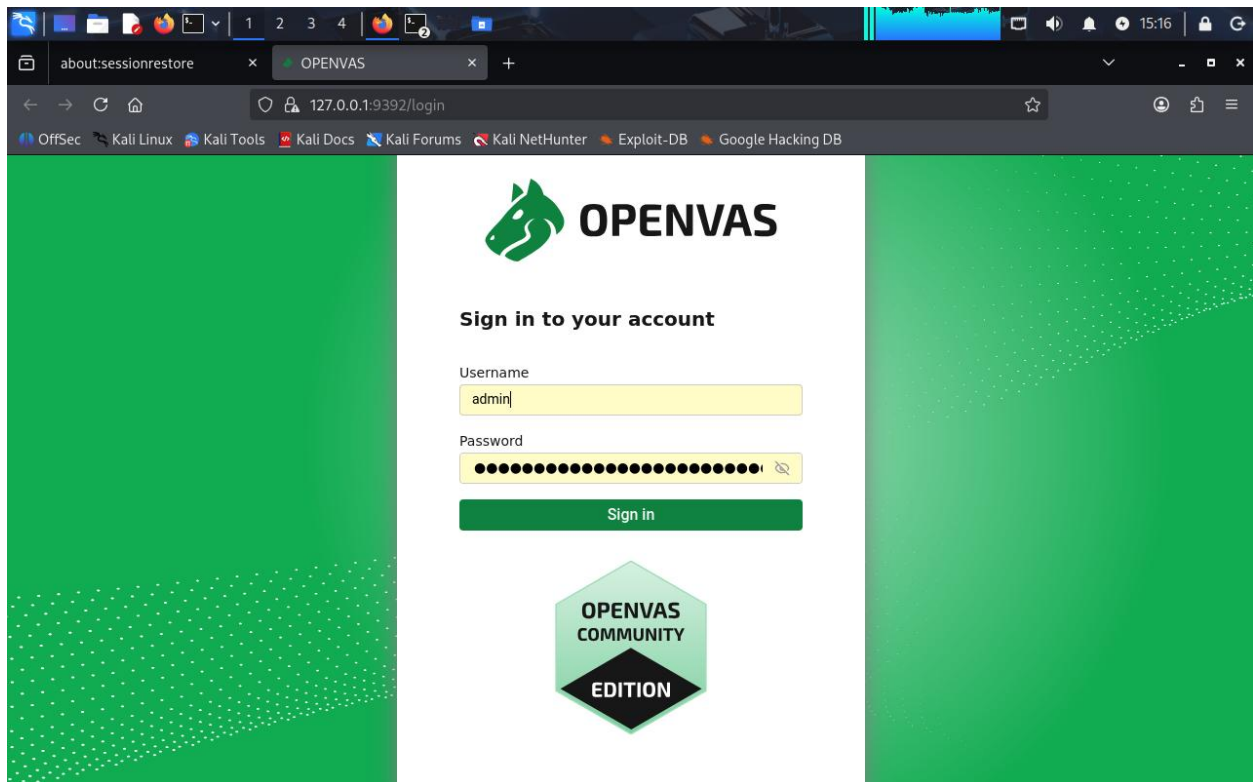
Excel or Google Sheets was used for documentation.

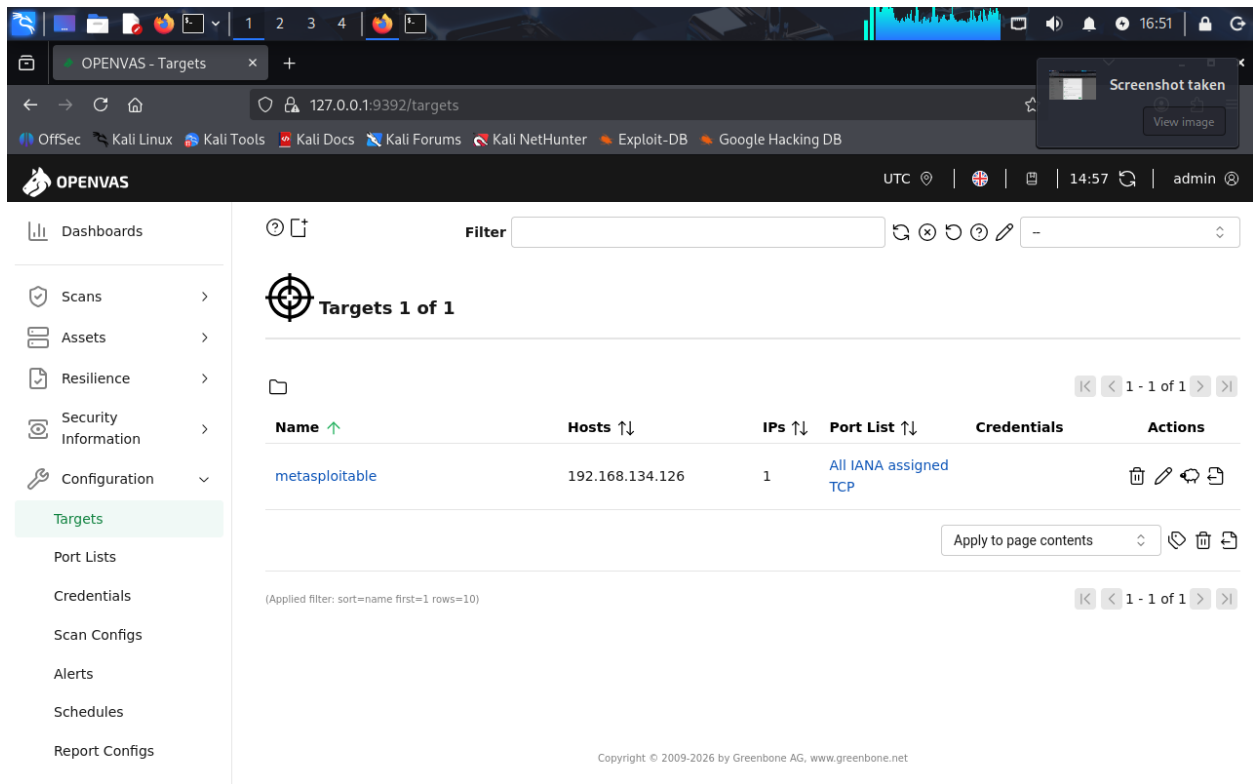
Environment Setup

Kali Linux and Metasploitable were installed in VirtualBox. Both virtual machines were connected on the same network so they could communicate with each other. Connectivity between both systems was verified before starting the assessment.

Vulnerability Scanning

OpenVAS was started from Kali Linux. After login, a new target was created using the IP address of the Metasploitable machine. A Full and Fast scan configuration was selected and the scan was started. The scanner checked the target for known vulnerabilities and misconfigurations.



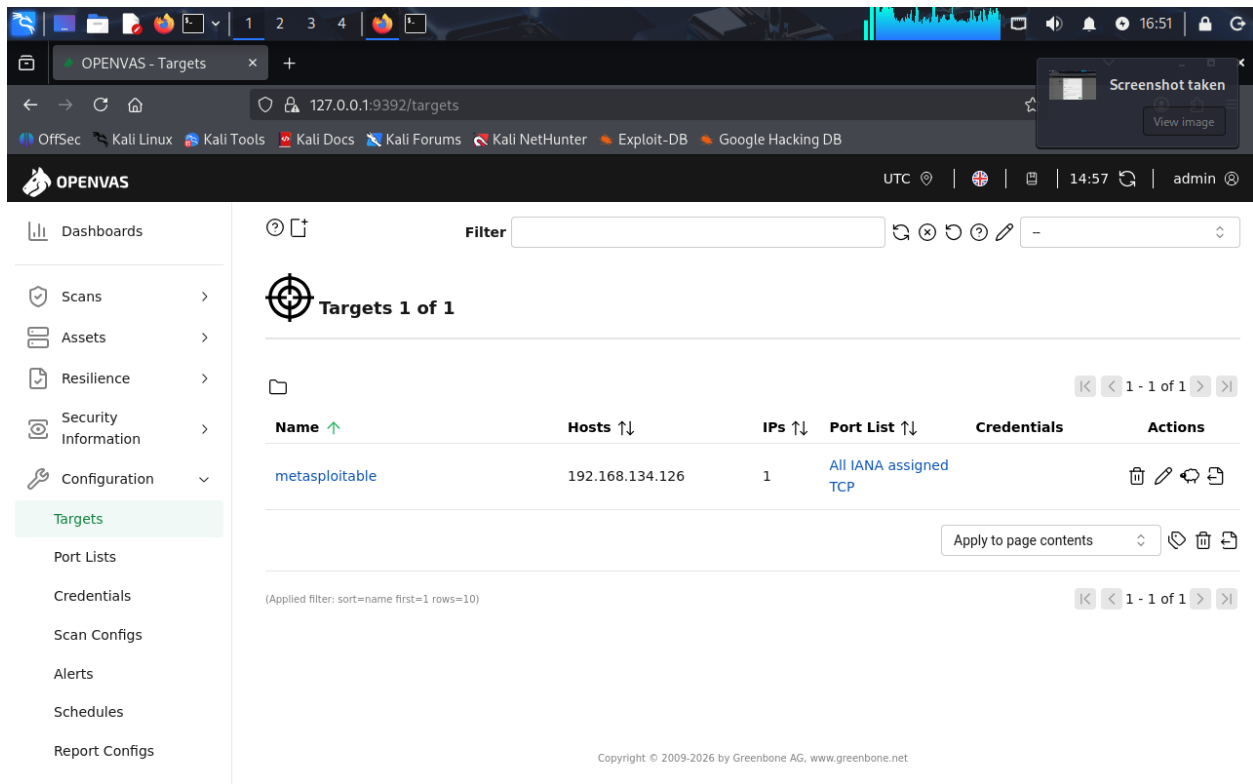


Nmap was also used to discover open ports and running services on the target machine.

Command used:

```
nmap -sV <Target-IP>
```

The scan identified services such as FTP, SSH, HTTP, and database ports.



Findings

The assessment identified multiple security issues such as open ports, outdated services, unnecessary exposed services, and weak configurations. These vulnerabilities could be used by attackers if left unpatched.

Examples of findings:

Open FTP service

Open SSH port

Web server running on HTTP

Outdated software versions

Greenbone Security Assistant - Mozilla Firefox

Kali Linux, an Offensive Security... x Greenbone Security Assistant x +

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Dashboard Scans Assets SecInfo Configuration Extras Administration Help

Task: Metasploitable 2 scan of IP 10.0.2.24

Modified: 11/24/2020 14:27:07 Owner: admin

Name: Metasploitable 2 scan of IP 10.0.2.24

Comment:

Target: Target for immediate scan of IP 10.0.2.24

Alerts:

Schedule: (Next due: over)

Add to Assets: yes

Apply Overrides: yes

Min QoD: 70%

Alterable Task: no

Auto Delete Reports: Do not automatically delete reports

Scanner: OpenVAS Default (Type: OpenVAS Scanner)

Scan Config: Full and fast

Order for target hosts: Sequential

Network Source Interface:

Maximum concurrently executed NVTs per host: 10

Maximum concurrently scanned hosts: 30

Status: Done

Duration of last scan:

Average scan duration:

Reports: 1 (Finished: 1, Last: Jul 24 2020)

Results: 135

Active Windows

Risk Assessment

The identified vulnerabilities were prioritized using CVSS scores and basic risk levels.

High Risk vulnerabilities were issues that could lead to serious compromise.

Medium Risk vulnerabilities had moderate impact.

Low Risk vulnerabilities were minor issues or informational findings.

Example:

Outdated service version – High Risk

Unnecessary open port – Medium Risk

Information disclosure – Low Risk

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Dashboard Scans Assets SecInfo Configuration Extras Administration Help

High N/A

1 3

Metasploitable 2 scan of ...

0 5 10 15 20

2

Done

Name	Status	Reports		Severity	Trend	Actions
		Total	Last			
Metasploitable 2 scan of IP 10.0.2.24	Done	1 (1)	Jul 24 2020	10.9 (High)		

1 - 2 of 2

Recommendations

Update all outdated services and software.

Close unused ports.

Disable unnecessary services.

Apply security patches regularly.

Use firewall rules for access control.

Perform regular vulnerability assessments.

Conclusion

This task provided practical understanding of how security assessments are performed using open-source tools. By using Kali Linux, OpenVAS, and Nmap, vulnerabilities were identified, risks were analyzed, and remediation steps were recommended.